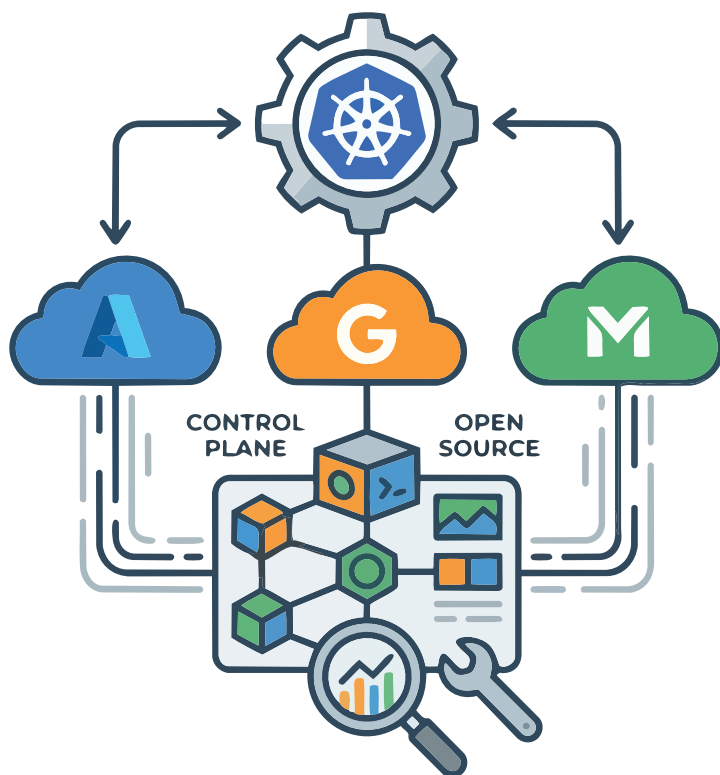


Managing Multi-Cloud Infrastructure: The Way Forward



Kubernetes and open source control planes make multi-cloud operations easier and help organisations build scalable and cloud-agnostic infrastructure platforms.

Crossplane, Cluster API, or GitOps, it becomes the basis to unify multi-cloud infrastructure management.

Understanding multi-cloud infrastructure

Using different cloud services from different companies for part or all your infrastructure is referred to as having a multi-cloud strategy. These infrastructures can consist of either public clouds, private clouds, or edge locations.

The benefits that come from having a multi-cloud infrastructure are numerous. These include the ability to:

- Minimise your risk of becoming tied down to one vendor (vendor lock-in);
- Improve your system's resiliency (the ability of your systems to bounce back) and recover from a disaster;
- Comply with government regulations around data residency;
- Utilise the location of workloads to provide better performance;
- Reduce infrastructure costs through better flexibility with providers.

The development and deployment of applications have changed dramatically with cloud computing. Over the last decade, cloud-based solutions have grown rapidly, providing businesses with a scalable, flexible, and fast way to innovate. Many enterprises today are increasingly adopting multi-cloud architectures, where applications and infrastructure are deployed across multiple cloud platforms. According to the Flexera State of the Cloud Report, 89% of organisations now operate in a multi-cloud environment, reflecting the growing demand for flexibility and resilience in cloud deployments. By

distributing workloads across different cloud providers, enterprises can reduce vendor lock-in, improve disaster recovery strategies, and meet regulatory or data-residency requirements. As cloud-native technologies continue to mature, multi-cloud infrastructure is becoming a key component of modern enterprise IT strategies.

Kubernetes and current open source control plane technologies are essential in this area of technology. They provide an orchestrated layer for containerised applications, enabling them to be portable from one cloud to another. When you use Kubernetes together with an open source control plane such as